

Artificial Intelligence Policy

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April 2025	April 2026	1.1

Purpose

Artificial Intelligence (AI) is the use of digital technology to create systems capable of performing tasks commonly thought to require human intelligence. UHP wishes to embrace and support the introduction of appropriately regulated and governed uses of AI related technologies that address a clinical or business need. As AI continues to take centre stage, the trust must make sure its models are fair, explainable, and secure. AI ethics and compliance frameworks will be the next frontier of digital Trust and will set the tone for how digital trust is maintained in the years ahead. The public have both an expectation and a fundamental right that personal information will be treated with care.

This policy sets out the way in which AI should be considered and used within the organisation.

Who should read this document?

Staff interested in the application of Artificial Intelligence or with the intent to implement a new solution which utilises Artificial Intelligence, prior to its use within the Trust.

Key Messages

- The Trust should be inquisitive and open to the new advancements in Artificial Intelligence and the potential efficiency and clinical benefits it can offer.
- Do not use any system or AI tool without prior consent from the Technical Design Authority (TDA) and relevant governance. This risk's exposing the trust to Cyber Security, Legal and Ethical issues regardless of the use case.
- The strategic alignment (both Clinical and Digital) of solutions must be considered by the relevant groups.

Core accountabilities

Owner	Chief Digital Officer (CDO) / Chief Technical Officer (CTO)
Consultation	Digital Sub Committee, D&I Senior Leadership Team, Digital Design Authority (Clinical + Technical), Imaging AI Lead, Information Governance, Business Informatics, Cyber Security, CxIO's
Ratification	Chief Digital Officer (CDO) / Chief Technology Officer (CTO)
Dissemination (Raising Awareness)	Communications Team
Compliance	Digital Design Authority (CDA & TDA)

Links to other policies and procedures

Version History

V1.0	February 2025	Initial draft for consultation
V1.1	April 2025	Feedback from IG, Imaging and Technical Design Authority incorporated, governance diagram enhanced. Signed off by Digital Sub Committee 01/05/2025

Equality, diversity, and inclusion is at the heart of the NHS strategy and for UHP this means ensuring we create an equitable organisation for colleagues, patients, and visitors. Our inclusion agenda recognises that everyone matters, as well as understanding that there are times when people, particularly those from protected groups (defined by the Equality Act 2010 - age; disability; gender reassignment; marriage and civil partnership; pregnancy and maternity; race; religion or belief; sex; sexual orientation), may face unfairness and discrimination.

We have a moral responsibility to work in a way that creates fairness as well as a legal and public sector equality duty which we take very seriously. We will always aim to create policies and practices that eliminate workplace discrimination and health inequalities, as well as strengthening our relationships within the community and with our colleagues who may identify with one or more protected groups.

A diverse workforce improves the experience of our patients and creates an environment where everyone belongs and feels included. UHP recognise that we all play a part in providing exceptional care to patients as well as showing kindness and compassion to patients and colleagues.

An electronic version of this document is available in the Document Library. Larger text, Braille and Audio versions can be made available upon request.

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1. Introduction

- 1.1. Artificial Intelligence (AI) is the use of digital technology to create systems capable of performing tasks commonly thought to require human intelligence. This policy is applicable to products, models, systems and technologies which use AI (including Machine Learning (ML)) and contain a component or function to enable or provide treatment or diagnosis or enable the management of health conditions for the purpose of health care.
- 1.2. This policy covers all applications of AI including software purchased or developed by the Trust that utilises AI as its primary or any secondary function. Primarily concerned with AI in Healthcare use cases; however, it is applicable to the use of AI in business process such as generational AI producing presentation slides, auto-rostering, recommendations in a system if fed from trust owned data (ML). These tools should be used with caution and under NHS recommended best practices and still adhere to all legal and local governance requirements laid out in this policy.

2. Purpose

- 2.1. This policy is applicable to all staff who wish to implement AI in any form into a system or process. The implementation and use of now widely available tools such as Chat GPT, Copilot etc when used in ways which do not have any application to healthcare or patient data follow pathway 1, all solutions with clinical uses, requiring hardware, access to patient identifiable/staff data or advice/prioritise on behalf of a healthcare professional follow pathway 2. AI is a changing landscape with new innovations into existing products and tooling, this policy is not exhaustive of all potential use cases and advice can be sought through [D&I IT Help](#) or the [CxlO team](#). All systems and tools must have their use registered through D&I onto the Trusts Information Asset Register.

Pathway 1: Generative AI Tools (e.g. email auto-response suggestions, software development auto-complete, copilot generative documents, knowledge base/incident management data ingestion).

Pathway 2: AI Applications in Healthcare (e.g. partnerships with AI startups for image analysis, integrating AI into processes to prioritise patient waiting lists, generative summary analysis of patient data).

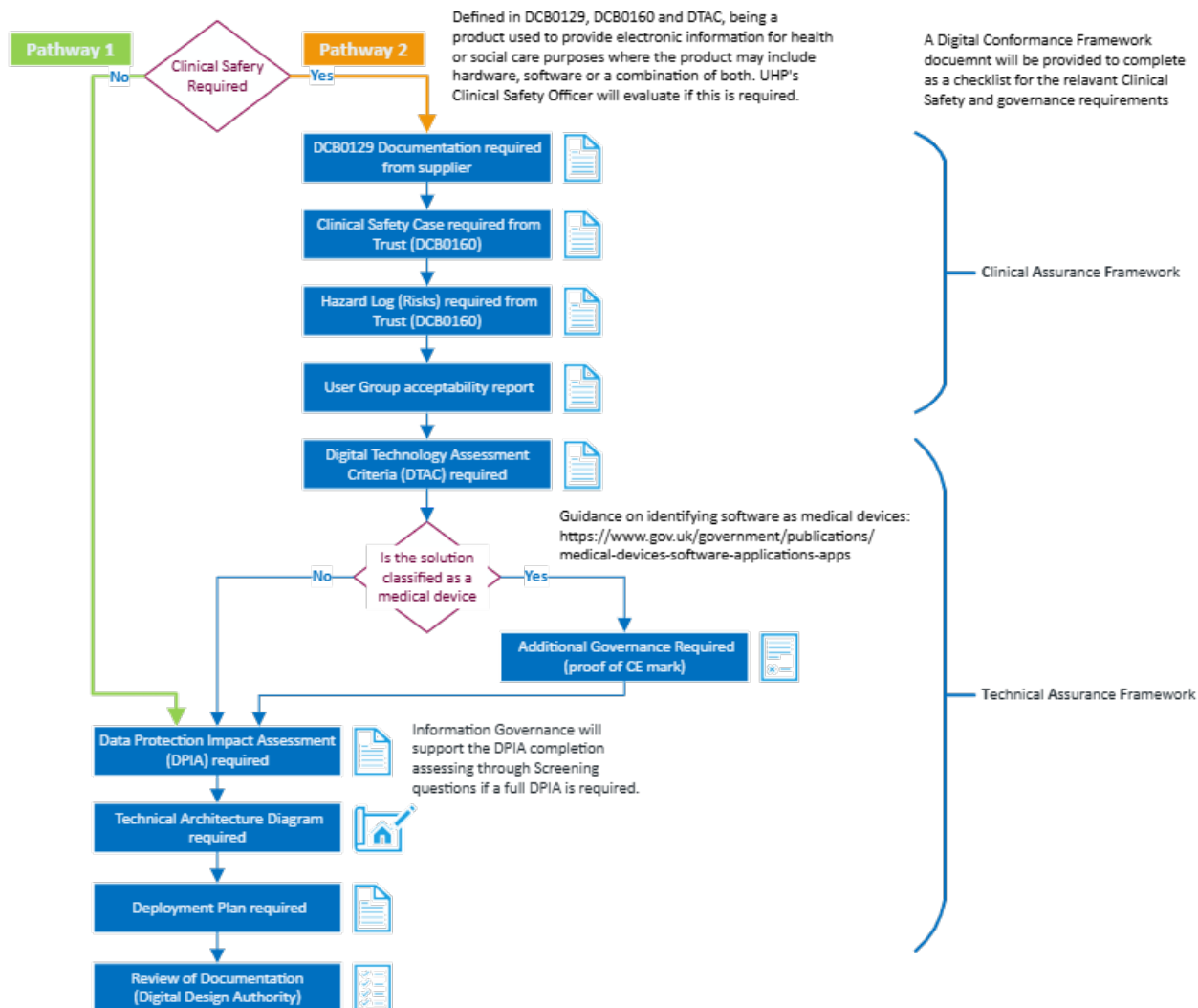


Figure 1 - Flowchart of Pathways for required governance (extract from full governance chart in Appendix 3)

2.2. The trust mandates any **Pathway 1** solution contains the following:

- The Trusts Information Governance (IG) lead, Data Protection Officer and Caldicott Guardian must be involved in any decision to implement or share data to develop AI technology.
- Suppliers produce a DTAC to provide assurance that technology meets minimum baseline standards (See Appendix 4)
- Gain approval to continue from the Technical Design Authority (TDA).

2.3. The trust mandates any **Pathway 2** solution contains the following:

- The Trusts IG lead, Data Protection Officer and Caldicott Guardian must be involved in any decision to implement or share data to develop AI technology.
- Provide a valid DCB0129 and DCB0160 safety case & hazard log.
- Suppliers produce a DTAC to provide assurance that technology meets minimum baseline standards (See Appendix 4)
- Gains approval to continue from the Clinical Design Authority (CDA) & Technical Design Authority (TDA).

- Recommended adherence to BS30440: This British Standard can serve as a guide for the safe and ethical development, implementation, and monitoring of AI products. Further, AI products can be assessed for conformity to this British Standard.

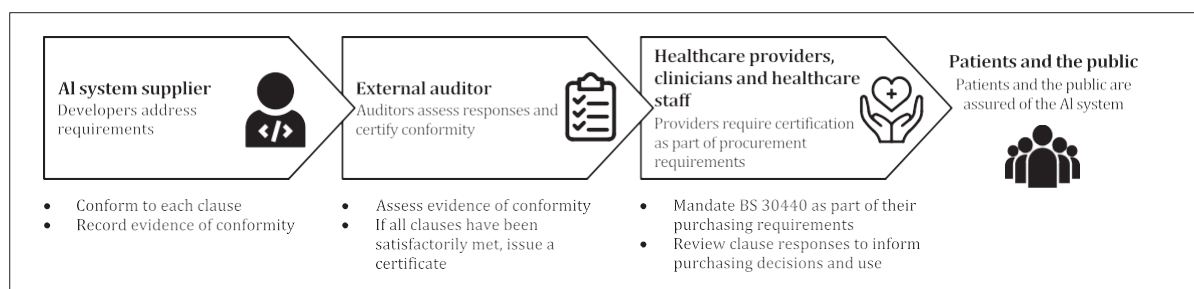


Figure 2 - BS30440 Summary

- 2.4. In practice, BS30440 British Standard could be used by several different parties involved in healthcare:
- 2.4.1. **AI system suppliers:** during the development of a given AI system, suppliers can choose to conform to the requirements in this British Standard, documenting responses and evidence against each clause for evaluation by external auditors.
 - 2.4.2. **External Auditors:** auditors can assess supplier responses and evidence against each clause, to determine whether the healthcare AI system conforms to this British Standard.
 - 2.4.3. **Healthcare Providers:** healthcare providers could mandate conformity to this British Standard as part of their purchasing requirements.
 - 2.4.4. **Clinicians and Healthcare Staff:** clinicians and staff at the procuring organisation can have confidence that healthcare AI products conforming to this British Standard are safe and effective.
 - 2.4.5. **Patients and the Public:** patients and the public can gain assurance that AI products used in the delivery of care have met the standards of quality and social equity required to achieve conformity to this British Standard; and
 - 2.4.6. **Other Stakeholders:** stakeholders such as pharmaceutical companies, healthcare insurers, innovation and market access consultancies and consumer technology companies.

3. Definitions

CDA - Clinical Design Authority. Provides clinical leadership for assuring digital developments are compliant with the Clinical Strategy, Clinical Safety and clinical pathways/processes.

Clinical Safety Case - A formal presentation of the Hazard Log, supported by a body of relevant evidence that provides a valid case that a system is safe for release.

DCB0129 - Clinical risk management requirements for manufacturers of Health IT Systems.

DCB0160 - Clinical risk management requirements for organisations that use Health IT Systems.

Digital Design Authority - Combined meeting of the Technical and Clinical Design Authorities.

DPIA - Data Protection Impact Assessment. A tool to analyse, identify and minimise risks to personal data.

DTAC - Digital Technology Assessment Criteria. A tool to assess suppliers at the point of procurement or as part of a due diligence process, to make sure digital technologies meet the NHS minimum baseline standard.

Generative AI - A form of AI that can create new content such as text, images, videos. Can learn from existing data and generate new data with similar characteristics. Examples include Microsoft Copilot and Chat GPT.

Hazard Log - A formal document which describes all the hazards that have been considered attributable to the product. It describes the hazards, potential clinical impact of those hazards, mitigations and controls and final assessment of risk.

Human-in-command - Capability to oversee the overall activity of the product (including its broader economic, societal, legal and ethical impact) and the ability to decide when and how to use the product.

Human-in-the-loop - Capability for human intervention in every decision cycle of the product.

Human-on-the-loop - Capability for human intervention during the design cycle of the system and monitoring the product's operation.

Supervised learning - Allows an AI model to learn from labelled training data.

TDA - Technical Design Authority. A decision-making body to provide technical authority, leadership and assurance to any proposed change or upgrade to validate against the Digital Strategy, technology blueprint and technology standards.

Reinforcement learning - Allows an AI model to learn as it performs a task.

Unsupervised learning - Training an AI algorithm to use unlabelled and unclassified information.

4. Duties

- 4.1. It is the responsibility of the **Technical Design Authority** (TDA) to review and assure a proposed AI solution is aligned strategically and the right 'technical fit' through cyber security, architecture and implementation. The TDA has the final decision on matters relating to Artificial Intelligence and may where appropriate can refuse a models use if it is deemed to not be compliant with any section of this, or national policy.
- 4.2. It is the responsibility of the **Clinical Design Authority** to review and assure a proposed AI solution used in Clinical use cases is aligned to the Clinical Strategy and has sufficiently mitigated risks, provided accurate supporting documentation for Clinical Safety.
- 4.3. **Information Governance Lead, Data Protection Officer** and **Caldicott Guardian** to assure AI solutions are used within legal precedents set and sufficient agreements and notices are in place.
- 4.4. Where Artificial Intelligence has been used, the creator, third party or Trust will need to ensure that clear accountable lines of responsibility and liability are available.
- 4.5. It is the responsibility of **All Staff** to adhere to this policy. Failure to comply may result in solutions actively blocked within the Trust network, expose protected data and have possible legal ramifications to the Trust and individual.

5. Overview of AI

- 5.1. Artificial intelligence (AI) has been adopted to automate tasks in many contexts, the application of AI technologies will result in a more personalised and efficient experience. For people working in the public sector, it means a reduction in the hours they spend on basic tasks, which will give them more time to spend on innovative ways to improve services.
- 5.2. AI is constantly evolving however it generally:
 - Involves machines using statistics to find patterns in large amounts of data.

- Is the ability to perform repetitive tasks with data without the need for constant human guidance.
- 5.3. Machine learning is a subset of AI, which can be:
- Supervised learning which allows an AI model to learn from labelled training data.
 - Unsupervised learning which is training an AI algorithm to use unlabelled and unclassified information.
 - Reinforcement learning which allows an AI model to learn as it performs a task.
- 5.4. AI products developed for a healthcare setting require rigorous evaluation so that they:
- Have demonstrable benefits.
 - Reach sufficient standards of safety and performance.
 - Successfully and safely integrate into the proposed environment.
 - Developed and used ethically.
 - Produce equitable outcomes.
- 5.5. The use of **Generative AI** is permitted where it does not breach GDPR, use data that would be deemed sensitive or confidential and does not bring the Trust into disrepute.
- 5.6. No member of the Trust should use patient or employee data with unapproved AI models. Specifically, patient data should never be uploaded to Generative AI such as Microsoft Copilot, Chat GPT or similar services.
- 5.7. Verify the Accuracy of Outputs - AI tools like Copilot are not perfect; they are constantly developing all the time, but this means that they can make mistakes and provide incorrect information. Outputs can:
- Contain factual inaccuracies or outdated information.
 - Feature mathematical errors, particularly in complex calculations – most AI tools are notably bad with any kind of maths.
 - Misinterpret or provide overly simplified responses.
- 5.8. Before using or sharing anything from Copilot or any AI tool, it is important that you understand the output yourself, and that you always review it thoroughly to ensure the accuracy and appropriateness of the content.
- 5.9. Be Mindful of Sources - Copilot may include links or references in its suggestions, but these are not always verified, and it has been known to make them up. When reviewing the source information:
- Check the reliability of the provided sources.
 - Cross-reference data with trusted/official platforms when needed.
- 5.10. **For advice on using Generative AI (including Microsoft Copilot and Chat GPT) please see Appendix 5.**
- 5.11. Understanding AI: [A guide to using artificial intelligence in the public sector - GOV.UK \(www.gov.uk\)](https://www.gov.uk/guidance/a-guide-to-using-artificial-intelligence-in-the-public-sector)

6. Assessing AI

- 6.1. For Pathway 2 (clinical use cases), prior to the introduction of any AI software, there must be evidence for a clinical need and current practice. This should be provided by the requesting body via internal audit to a template which will be used for further pre and post deployment Quality Assurance (QA).

- 6.2. With an AI project you should consider several factors, including AI ethics and safety. These factors span safety, ethical, legal and administrative concerns and include:
- **Data Quality** - the success of your AI project depends on the quality of your data.
 - **Fairness** - are the models trained and tested on relevant, accurate, and generalisable datasets and is the AI system deployed by users trained to implement them responsibly and without bias.
 - **Accountability** - consider who is responsible for each element of the model's output and how the designers and implementers of AI systems will be held accountable.
 - **Privacy** - complying with appropriate data policies, for example the General Data Protection Regulations (GDPR) and the Data Protection Act 2018.
 - **Explainability and transparency** - so the affected stakeholders can know how the AI model reached its decision.
 - **Costs** - consider how much it will cost to build, run and maintain an AI infrastructure, train and educate staff and if the work to install AI may outweigh any potential savings.
 - **Transparency** - consider how you will engage and address concerns from staff, patients and public around the use of their data, the requirement to offer an opt-out, and how you will measure the reliability and who has accountability.
- 6.3. **Third party applications**
- 6.4. All products using AI must have the ability to "turn off" this function (unless the system is purely for AI, in which case the process must have redundancy built in that prevents a loss of service should the tool become unavailable for any reason).
- 6.5. **No uses of AI are permitted to trial or use without approval from the Technical Design Authority (TDA).**
- 6.6. All third-party applications that use AI must be registered on the Trust's Information Asset Register and reviewed yearly. Providing a named System Manager and Deputy System Manager, responsible for the administration and compliance of the system including annual System Assurance Toolkit).
- 6.7. All third-party applications must provide an up-to-date contact address should a member of the public have concerns or questions.
- 6.8. All third-party applications must disclose the type of model used so that the accuracy of their model can be fairly assessed against its competitors.
- 6.9. Where a third-party application is intending to use Trust data to train their model, they must disclose the data used and be able to rationalise the reason for its incorporation.
- 6.10. **What information should you ask of a supplier?**
- 6.10.1. For Pathway 2 (clinical use cases) the recommendation is adherence to the BS30440 standard. A framework which the supplier can be externally audited against. Like DCB0129 there are specific files a supplier should be able to supply on request which form evidence of supplier due diligence and conformity to the standard as well as Risks, Bias's and maintenance of the system.
- 6.11. **For detailed guidance on the relevant factors and impacts to assess an AI innovation please refer to Appendix 6: Key Excerpts from BS30440:2023.**
- 6.12. [Assessing if artificial intelligence is the right solution - GOV.UK \(www.gov.uk\)](https://www.gov.uk/government/publications/assessing-if-artificial-intelligence-is-the-right-solution)

7. Planning for AI Implementation

- 7.1. All AI use cases must have human oversight in the decision-making process (Human-in-the-loop).

- 7.2. All uses of AI must, where relevant, have processes in place where a human can intervene if a model produces incorrect, harmful, or misleading information. Known as Human-in-the-loop, no automated decisions can be made that could impact a patient's health care without adequate review from a person of equivalent authority to that of the non-AI process.
- 7.3. It is noted that third parties will require human intervention in their terms of service for any of its AI products, however, when assessing incidents, liability may be attributed as follows:
- 7.3.1. In the case that the Artificial Intelligence model was developed by a third party and was used in accordance with the third parties' recommendations and Trust Policy then the liability is with the third party.
- 7.3.2. In the case that the Artificial Intelligence model was developed by a third party and was **not** used in accordance with the third parties' recommendations and Trust Policy then the liability is with the user.
- 7.3.3. In the case that the Artificial Intelligence model was developed internally then the liability is with the organisation.
- 7.3.4. Any potential discrepancy (that results in clinical harm) between AI interpretation and the final human generated report should undergo the same duty of candour process as any other non-AI related discrepancy.
- 7.3.5. Comments of our Insurer regarding legal standing: "On a strict interpretation of the regulations governing CNST, cover can only be engaged in respect of an *"act of omission on the part of...a person employed or engaged by the Trust"*. Whilst it is recognised that part of the value of AI is to remove the individual from the process thereby freeing up resources, I think it follows that there will have to have been a person involved somewhere along the line (indeed, even if only in respect of the decision to use AI in the first place). To the extent, therefore, that it can be alleged that a person employed by the Trust acted in breach of duty, then CNST cover will be in place."
- 7.4. All uses of AI must have contact information that is dedicated to allowing the users to report issues or concerns about the model's behaviour. This must be a mailbox which is monitored frequently to ensure timely responses to concerns.
- 7.5. There must be a responsible clinician designated by the requesting body who will oversee the collection and analysis of ongoing quality assurance (which should be quarterly in the first instance).
- 7.6. Following successful procurement, the analysis and performance of the training cohort used for the software must be made available to the responsible clinician/clinical AI lead to act as baseline performance data.
- 7.7. Following satisfactory performance of the training cohort, there should be a period of shadow testing (parallel analysis on live cases) with results only available to the responsible clinician / designated small group of users. Satisfactory audit of the shadow testing must be performed and any subsequent performance issues corrected before any live use can be considered.
- 7.8. There is rarely a reason to use patient identifiable information for the training of a machine learning model. To reduce the risk of accidental disclosure, the following information should not be used in a training dataset.
- Unique Identifiers (MRN, NHS Number, Hospital Number).
 - Name.
 - DOB (Age or age bandings can be used).
 - Full Line Address (Postcode sectors can be used).
- 7.9. Exceptions to this can only be granted through the CDA & TDA where possible using the highest level of anonymisation.

- Synthetic Data (highest, but least accurate).
- Anonymised Data (data cannot be reidentified).
- Pseudonymised (data can be reidentified only with the correct key).
- Patient Identifiable Data (lowest).

- 7.10. For any AI software being used for clinical care, there should be considerations by the developer to ensure that the results can be explained in layperson terms to aid clinical professionals in relating technical information to patients.
- 7.11. No Artificial Intelligence tool is permitted to delete or alter any clinical or legally binding document including patients' medical notes and contracts. Such tools should only be permitted to add information, and in the case of amendments must be reviewed by a professional before overwriting (Human-in-the-loop).
- 7.12. Any Artificial Intelligence model used in clinical applications must be extensively tested to ensure accurate predictions across a wide range of demographic groups. For example, in the case of skin image analysis, the full range of Fitzpatrick skin types should be used. It is recommended that any results are independently reviewed by a suitably qualified subject matter expert.
- 7.13. **Predictive & Prescriptive Analytics**
- 7.13.1. Predictive & Prescriptive Analytics covers a wide range of Data Science and Operational Research methodologies, including but not limited to classification, regression, clustering, anomaly detection, simulation modelling, deep learning, neural networks, forecasting, natural language processing, sentiment analysis.
- 7.13.2. Predictive and Prescriptive techniques should only be performed by trained professionals.
- 7.13.3. All Artificial Intelligence models developed or modified by the Trust must have full version control providing an audit trail and accountability (e.g. via Git).
- 7.13.4. Artificial Intelligence applications that become part of critical infrastructure or extend use beyond original approved scope must be reapproved by the CDA & TDA.
- 7.13.5. All applications of AI within the Trust, where they are part of a critical business function, must have redundancy built in that prevents a loss of service should the tool become unavailable for any reason. This should be detailed in the Trust Business Continuity Plan.
- 7.14. [Planning and preparing for artificial intelligence implementation - GOV.UK](#)

8. Managing AI Projects

- 8.1. Projects requiring D&I resource to implement should submit a project request in the first instance. [D&I Project Request - University Hospitals Plymouth NHS Trust - ITSM](#)
- 8.2. For projects which do not require D&I resource but are implementing AI or using a new third-party tool, the governance listed in this policy must still be followed. Requests for TDA & CDA review can be made via IT Help 'Request Something'. Failure to have the correct approval may result in solutions actively blocked within the Trust network, expose protected data and have possible legal ramifications to the Trust and individual. Full governance requirements are detailed in Appendix 3.
- 8.3. [Managing your artificial intelligence project - GOV.UK \(www.gov.uk\)](#)

9. AI Clinical Safety & Governance

- 9.1. Information Governance should be advised of all proposed changes involving the use of AI and the necessary Data Sharing Agreements and privacy notices required.
- 9.2. Any system (AI included) with the potential to cause harm to patients, for example (but not limited to): system failures, algorithmic flaws or misuse, must be compliant with the

established framework of clinical risk management - DCB0160 that was incorporated into English law in section 250 of the Health and Social Care Act 2012.

- 9.3. A third-party company working with Software and AI and under the remit of DCB0129 must have completed and had approved a Clinical Safety Case and Hazard Log as part of the clinical risk management framework before the commencement of any work.
- 9.4. A third-party company working with Software and AI must have completed and had approved a Digital Technology Assessment Criteria (DTAC) before the commencement of any work.
- 9.5. A third-party company working with Software and AI must have completed and had approved a Data Protection Impact Assessment (DPIA) before the commencement of any work.
- 9.6. It is further recommended that any application of AI adheres to British Standard BS30440 to serve as a guide for the safe and ethical development, implementation and monitoring of AI products, and to holds suppliers accountable to the conformity of the standard.
- 9.7. Any new AI asset will require registration with the Trust's Information Asset Register through D&I.
- 9.8. It is not permitted to use Artificial Intelligence to track the physical location of any patient, employee, or member of the public without their informed consent, unless done so with legal precedent and Trust approval (e.g. in response to security event or to prevent harm where the use case has been reviewed and pre-approved).
- 9.9. The use of software that trains on a person's likeness, such as identifying a person's identity from image or video recognition software is not permitted without authorisation by the Trust's TDA.
- 9.10. Artificial Intelligence can be exploited to automate and amplify harassment through chatbots, deepfakes, and social engineering, making it easier to spread harmful content, impersonate individuals, or orchestrate coordinated attacks to our critical infrastructure (software/estates/robotics). Alongside this, the availability of AI enhanced methodologies creates easier access to malicious software and increases the success of scams. Whilst AI can be used to improve the protections of anti-virus and detection, colleagues should be prepared to prevent misuse where possible.
- 9.11. The threat of technologies like deepfakes and generative AI increases the capabilities of threat actors to create realistic fake personas, impersonate real people and create convincing messages. This increases the likelihood of security risks, such as phishing, fraudulent communications, data breaches and impersonating or faking patients and consent. Therefore, it is not permitted to use any AI model that recreates the likeness or can be used to impersonate a member of staff.
- 9.12. You should not use Artificial Intelligence Technologies that use unauthorised data or are not transparent about the source of its data for its training set.
- 9.13. All Artificial Intelligence models that have been classified as Software and AI as a Medical Device must be compliant with the Medical Device Regulations 2002 (UK MDR 2002) and any additional Medicines & Healthcare Regulatory (MHRA) recommendations including but not limited to Software and AI as a Medical Device Change Programme WP9-01 to 11.
- 9.14. All the current legal acts not listed in this policy by pertaining to the application of AI still apply and should be considered when using or designing AI.

10. AI Ethics & Bias

- 10.1. The Trust is committed to ensuring that the use of AI in our organisation supports equality and inclusion for all individuals. This policy aims to prevent bias and discrimination in AI systems and to promote fairness and transparency in decision-making processes.
- 10.2. AI systems should be designed and implemented in a way that prevents unlawful discrimination based on protected characteristics as defined by the Equality Act 2010.

These include age, disability, gender reassignment, marriage and civil partnership, pregnancy and maternity, race, religion or belief, sex, and sexual orientation.

- 10.3. AI models used or designed by the organisation must be thoroughly tested to ensure they do not introduce or perpetuate bias against any group.
- 10.4. Developers and third-party suppliers are required to demonstrate that AI systems are designed using diverse data representative of the population of the Trust, including anyone who receives healthcare but does not reside in within the catchment area to minimise potential bias (e.g. don't train AI models solely with data from inner-city New-York and expect it to be representative of Devon & Cornwall).
- 10.5. AI systems trained on existing data may be implicitly or explicitly unbalanced by either under or over representing characteristics. This can also be true for the creator and third parties developing AI models who may have preconceived perceptions of said data patterns and inadvertently sway the model in a way that leads to discrimination. The organisation has a responsibility to ensure that any AI system in use is not unlawfully discriminatory under the Equality Act 2010 and the Data Protection Act 2018. Compliance with one does not guarantee compliance with the other.
- 10.6. All individuals must have equal access to the services and opportunities provided by AI systems. AI should not create barriers or restrict access based on discriminatory factors.
- 10.7. Automated decisions made by AI **must be subject to human review** (Human-in-the-loop), particularly when they may significantly affect an individual's rights, healthcare, or access to services.
- 10.8. Processing of Personal Data must be 'Fair.' Any processing of personal data using AI that leads to unjust discrimination of people, will violate the fairness principle.
- 10.9. Data protection aims to protect individuals' rights and freedoms, which includes the right to not be discriminated against. The organisation is responsible for ensuring reasonable technical and organisational measures are in place to prevent this.
- 10.10. The processing of personal data for profiling and any automated decision-making may lead to discrimination, the organisation is responsible for ensuring reasonable technical and organisational measures are in place to prevent this.
- 10.11. Article 22 of the GDPR states that the **data subject shall have the right not to be subject to a decision based solely on automated processing**, including profiling which produces legal effects concerning him or her or similarly significantly affects them.
- 10.12. [Understanding artificial intelligence ethics and safety - GOV.UK \(www.gov.uk\)](https://www.gov.uk/government/publications/understanding-artificial-intelligence-ethics-and-safety)

11. Overall Responsibility for the Document

- 11.1. The overall responsibility for this policy is the Chief Technology Officer (CTO) with the decision of any AI application across the trust made by the Digital Design Authority (combination of TDA & CDA).

12. Consultation and Ratification

- 12.1. The design and process of review and revision of this policy will comply with The Development and Management of Formal Documents.
- 12.2. The review period for this document is set as default of 12-months from the date it was last ratified, or earlier if developments within or external to the Trust indicate the need for a significant revision to the procedures described.
- 12.3. This document will be reviewed by the Digital Design Authority & CIGAC and ratified by the Chief Technology Officer (CTO).
- 12.4. Non-significant amendments to this document may be made, under delegated authority from the Chief Technology Officer (CTO), by the nominated owner. These must be ratified by the Chief Technology Officer (CTO).

- 12.5. Significant reviews and revisions to this document will include a consultation with named groups, or grades across the Trust. For non-significant amendments, informal consultation will be restricted to named groups, or grades who are directly affected by the proposed changes.

13. Dissemination and Implementation

- 13.1. Following approval and ratification, this policy will be published in the Trust's formal documents library and all staff will be notified through the Trust's normal notification process.
- 13.2. Document control arrangements will be in accordance with The Development and Management of Formal Documents.
- 13.3. The document owner will be responsible for agreeing the training requirements associated with the newly ratified document with the named Chief Technology Officer (CTO) and for working with the Trust's training function, if required, to arrange for the required training to be delivered.
- 13.4. The latest version of this document will be held in the Trusts digital document library accessible to all staff.

14. Monitoring Compliance and Effectiveness

- 14.1. Any implementation of AI should be regularly reviewed for its appropriateness (recommended minimum yearly).
- 14.2. An AI model should be assessed for how it preforms against unforeseen circumstances and data. The best evaluation metric will depend on the problem you are trying to solve, and your chosen model. You should also consider the ethical, economical and societal implications. These considerations make the fine tuning of AI systems relevant to both data scientists and delivery leads.
- 14.3. Considerations for AI models will include:
- What level of performance your problem needs.
 - How interpretable you need your model to be.
 - How frequently you need predictions or retraining.
 - The cost of maintaining the model.
- 14.4. The hazard log should also be regularly updated with any new risks or hazards as they are identified or as new mitigations become possible. Hazards with a considerable risk or likelihood should also be recorded on the Trusts Risk Register.
- 14.4.1. Following rounds of post deployment quality assurance, the responsible clinician must provide data to prove that the software is cost-effective and actively making a positive difference in clinical care.
- 14.5. Applications of AI once approved by the TDA or CDA may be asked to present updates on progress or findings back to the groups for shared learning.

15. References and Associated Documentation

- 15.1. This policy was developed with thanks to Somerset NHS Foundation Trust for providing their AI Policy for comparison and alignment.
- 15.2. [BSI BSOL Standards Online BS 30440:2023](#)
- 15.3. Data compliance checklist for adopters:**

- 15.3.1. [AI and Digital Regulations Service for health and social care - AI regulation service - NHS \(innovation.nhs.uk\)](https://innovation.nhs.uk)
- 15.3.2. [Understanding the evidence for a technology before adopting it - AI regulation service - NHS \(innovation.nhs.uk\)](https://innovation.nhs.uk)
- 15.3.3. [Data compliance checklist for adopters - AI regulation service - NHS \(innovation.nhs.uk\)](https://innovation.nhs.uk)
- 15.3.4. [A guide to using artificial intelligence in the public sector - GOV.UK \(www.gov.uk\)](https://www.gov.uk)
- 15.3.5. [The NHS AI Lab - NHS Transformation Directorate \(england.nhs.uk\)](https://england.nhs.uk)

Appendix 1: Dissemination Plan and Review Checklist

Dissemination Plan			
Document Title		Artificial Intelligence Policy	
Date Finalised		April 2025	
Previous Documents			
Action to retrieve old copies		N/A	
Dissemination Plan			
Recipient(s)	When	How	Responsibility
All Trust staff	Once policy added to document library.	Digital Design Authority, Information Governance, StaffHub Page	Communications Team

Review Checklist		
Title	Is the title clear and unambiguous?	Yes
	Is it clear whether the document is a policy, procedure, protocol, framework, APN or SOP?	Yes
	Does the style & format comply?	Yes
Rationale	Are reasons for development of the document stated?	Yes
Development Process	Is the method described in brief?	Yes
	Are people involved in the development identified?	Yes
	Has a reasonable attempt has been made to ensure relevant expertise has been used?	Yes
	Is there evidence of consultation with stakeholders and users?	Yes
Content	Is the objective of the document clear?	Yes
	Is the target population clear and unambiguous?	Yes
	Are the intended outcomes described?	Yes
	Are the statements clear and unambiguous?	Yes
Evidence Base	Is the type of evidence to support the document identified explicitly?	Yes
	Are key references cited and in full?	Yes
	Are supporting documents referenced?	Yes
Approval	Does the document identify which committee/group will review it?	Yes
	If appropriate have the joint Human Resources/staff side committee (or equivalent) approved the document?	N/A
	Does the document identify which Executive Director will ratify it?	Yes
Dissemination & Implementation	Is there an outline/plan to identify how this will be done?	Yes
	Does the plan include the necessary training/support to ensure compliance?	Yes
Document Control	Does the document identify where it will be held?	Yes
	Have archiving arrangements for superseded documents been addressed?	Yes
Monitoring Compliance & Effectiveness	Are there measurable standards or KPIs to support the monitoring of compliance with and effectiveness of the document?	Yes
	Is there a plan to review or audit compliance with the document?	Yes
Review Date	Is the review date identified?	Yes
	Is the frequency of review identified? If so is it acceptable?	Yes
Overall Responsibility	Is it clear who will be responsible for co-ordinating the dissemination, implementation and review of the document?	Yes

Appendix 2: Equalities and Human Rights Impact Assessment

Core Information

Date	April 2025
Title	Artificial Intelligence Policy
What are the aims, objectives & projected outcomes?	To ensure the ethical, fair, and secure implementation of AI technologies within the Trust. The policy seeks to promote equality and diversity, eliminate discrimination, and safeguard personal information. The projected outcomes include enhanced clinical and business efficiencies, improved patient care, and the establishment of robust governance frameworks to maintain digital trust. By adhering to these principles, the Trust aims to create an inclusive environment that respects the rights and dignity of all individuals, ensuring equitable access to AI-driven services and preventing bias in AI applications

Scope of the assessment

Evaluating the potential impacts of AI implementation on various protected characteristics and human rights. This involves assessing how AI technologies might affect individuals based on race, religion, disability, sex, gender identity, sexual orientation, age, and socio-economic status. Internal stakeholders have been consulted across various areas from the organisation, with external stakeholders both providing comparable policies and evaluating this document. This policy is available in all forms of communication upon request and contains no restriction or prejudice to any group, specifically preventing such within solutions.

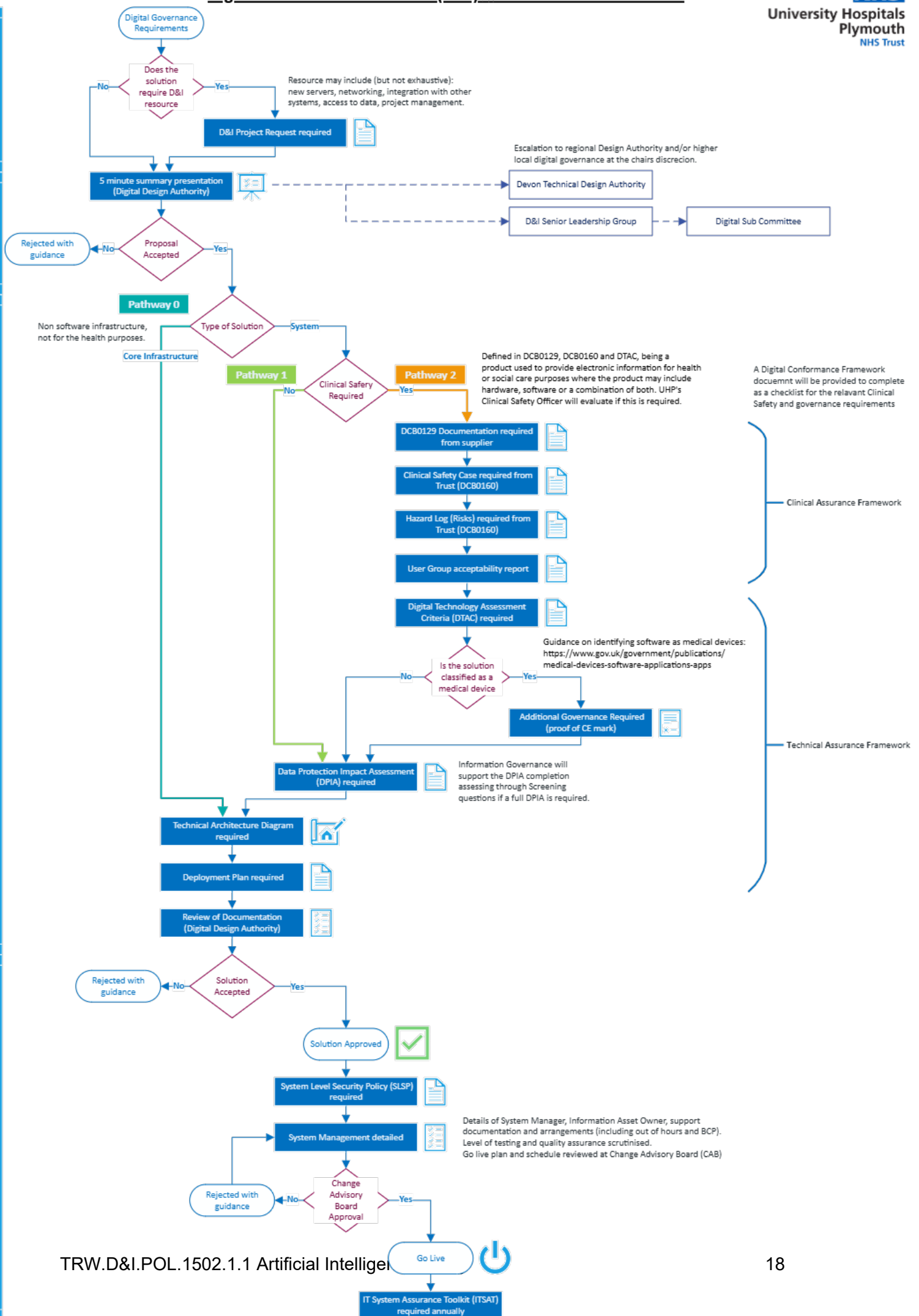
Collecting data

Race	This is mitigated as the policy can be made available in alternative languages.
Religion	The document has no impact in this area.
Disability	This is mitigated as the policy can be made available in alternative formats.
Sex	The document has no impact in this area.
Gender Identity	The document has no impact in this area.
Sexual Orientation	The document has no impact in this area.
Age	The document has no impact in this area.
Socio-Economic	The document has no impact in this area.
Human Rights	The document has no impact in this area.
What are the overall trends/patterns in the above data?	There are no trends/patterns in this data.
Specific issues and data gaps that may need to be addressed through consultation or further research	Trust wide documents can be made available in a number of different formats and languages if requested.

Involving and consulting stakeholders

Internal involvement and consultation	Digital Sub Committee D&I Senior Leadership / Heads of Department Technical Design Authority D&I Service Management Improvement Team CxIO Team, Information Governance, Imaging AI Lead, Business Informatics, Cyber Security			
External involvement and consultation	Torbay and South Devon NHS Foundation Trust Programme Director for Digital Innovation and Transformation. Somerset NHS Foundation Trust.			
Impact Assessment				
Overall assessment and analysis of the evidence	No impact to current processes. Once publicised staff and patients may understand more about the use of their data in automated processes but with clear explanation of their right to opt out from fully automated processes, and that no trust solution shall be fully automated with AI without Human-in-the-loop.			
Action Plan				
Action	Owner	Risks	Completion Date	Progress update
Publicise and promote the policy and Digital Design Authority to all staff	Digital Design Authority	-	Q2 2025	
Simplify access to D&I governance and approval through ServiceNow (IT Help)	D&I	-	Q2 2025	
Collect full register of AI systems currently in place	D&I Enterprise Architecture	Relies on responses from end users	Q3 2025	
Public/Patient facing promotion of AI guidance, policy and use cases	Digital Design Authority		Q3 2025	

Appendix 3: Full D&I Governance Requirements



Appendix 4: Digital Technology Assessment Criteria (DTAC)

As outlined from NHSE for best practice, the NHS DTAC sets out specific standards for healthcare technologies. If you want your technology to be used in the NHS and social care, it needs to meet these standards.

During procurement, adopters will review your completed DTAC to assess if your healthcare technology meets minimum baseline standards. If your technology meets the standards, adopters feel assured they are buying a safe and effective technology. They are then more likely to buy your technology.

Suppliers are to provide a DTAC that focuses on 5 core areas:

- clinical safety
- data protection
- technical assurance
- interoperability
- usability and accessibility

The DTAC brings together legal requirements and best practice in these areas. It overlaps with other legal requirements, such as conformity with medical device or data protection regulation.

[Digital Technology Assessment Criteria \(DTAC\) - Key tools and information - NHS Transformation Directorate](#)

Appendix 5: Advice for using Generative AI

AI (Artificial Intelligence) is something that many of you will now be aware of, especially as tools which can answer questions or generate content, such as Microsoft Copilot and ChatGPT.

AI tools like Microsoft Copilot within Microsoft Edge provide innovative ways to assist users with day-to-day tasks, such as summarising the points on a page. Many staff may have noticed and started using this tool. However, while these AI tools can be helpful, it's essential to exercise caution when using them as they can sometimes provide inaccurate information depending on the way the tool interrogates your prompt.

Below are some key points to consider when using Microsoft Copilot—or any AI software—at work (**please note:** some specific uses of more advanced AI tooling have been approved for use in health care settings, this guidance only relates to AI for generating content, not for direct Healthcare use):

Never Use Patient, Staff or Sensitive Information

Microsoft Copilot operates in the cloud and uses large language models to process your requests. This means that the information you input could potentially be stored or used to improve the AI. **Therefore, we would ask staff to avoid entering any staff, patient, or sensitive data.** Keeping our data secure is of vital importance, and sharing sensitive information with AI applications could result in data breaches or non-compliance with data protection regulations.

Verify the Accuracy of Outputs

AI tools like Copilot are not perfect; they are constantly developing all the time, but this means that they can make mistakes and provide incorrect information. Outputs can:

- Contain factual inaccuracies or outdated information.
- Feature mathematical errors, particularly in complex calculations – most AI tools are notably bad with any kind of maths.
- Misinterpret or provide overly simplified responses.

Before using or sharing anything from Copilot or any AI tool, it is important that you understand the output yourself, and that you always review it thoroughly to ensure the accuracy and appropriateness of the content.

Be Mindful of Sources

Copilot may include links or references in its suggestions, but these are not always verified, and it has been known to make them up. When reviewing the source information:

- Check the reliability of the provided sources.
- Cross-reference data with trusted/official platforms when needed.

Stay Informed About Policies

Ensure you are familiar with the Trust's policies for Internet Use, Information Security and Artificial Intelligence.

If you need help...

AI tools like Microsoft Copilot can enhance productivity, but their use requires caution. While the tools can save time, it's essential to remember that AI cannot replace human judgment or expertise. By being cautious, verifying the accuracy of any outputs, and not inputting sensitive information, we are able to use AI responsibly and effectively.

Please be aware that currently we are not able to provide licences for the M365 Copilot app, but the free version is currently enabled in Microsoft Edge. If you have any questions or concerns about using AI tools like Copilot, please reach out to the M365 Support Team at plh-tr.m365supportteam@nhs.net

Appendix 6: Key Excerpts from BS30440:2023

Stakeholder involvement

- The supplier shall demonstrate that the product addresses a healthcare need through:
 - a) objective assessment: published research on the ongoing impact of the current healthcare need in the context of existing good practice solutions; and
 - b) subjective assessment: evidence that relevant stakeholders (e.g. health providers, patients and/or the public) agree that the identified need exists.

EXAMPLE

1. Objective assessment: having identified that the need exists and explored possible AI solutions, the supplier undertakes a review of existing good practice solutions, including published research on the impact of the current healthcare need, such as:
 - I. detrimental effect on quality of life or life expectancy or healthcare safety;
 - II. economic impact; and
 - III. operational burden to healthcare pathways, e.g. workflow studies;
2. Subjective assessment: the supplier is considering the development of a product that utilises imaging and healthcare data to predict and diagnose the recurrence of cancer in patients in remission undergoing routine surveillance. The organisation conducts a series of interviews, focus groups and surveys with radiologists, oncologists, nurse specialists and patients to establish that the need exists, and how to best meet it.

(BS30440:2023, Section 4)

- The supplier shall document how stakeholders were identified and provide a comprehensive record of all stakeholder engagement during development of the product (BS30440:2023, Section 5.1.2).
- The supplier shall provide evidence that the development of the product reflects good practice standards for usability (BS30440:2023, Section 5.1.3).

NOTE: BS EN 62366-1 provides requirements for application of usability engineering to medical devices.

Training Data

- The supplier should demonstrate that training data was obtained and handled ethically, providing summary statistics of its characteristics (BS30440:2023, Section 5.2).
 - The supplier shall provide an illustration of the flow of training data from where, when, and how it is generated through to its use in the development of the product.

EXAMPLE

If a radiographer acquires data and a data officer curates and transfers data for analysis, separate touchpoints can be included.

Ethics and governance

- The supplier shall document the ethical and legal frameworks that govern data access and use for the purpose of developing the product (BS30440:2023, Section 5.2.2.1).
- In situations where the framework does not require it, the supplier shall state their understanding as to whether individuals:
 - c) were informed about the use of their personal data; and
 - d) consented to the use of their personal data for the development of the product and if so, how consent was requested and documented.

(BS30440:2023, Section 5.2.2.3)

- The supplier shall document:
 - the purpose for which the data were originally collected;
 - whether the purpose was specifically for the development of the product;
 - the context in which they were collected; and
 - their relevance to the purpose and context of intended use.

EXAMPLE

Data could have been originally collected for a research ethics-approved study or from routine health and care-related interactions, each of which are examples of original collection purpose. The religious, geographical or cultural settings from which data were collected are examples of the collection context.

(BS30440:2023, Section 5.2.2.4)

- The supplier shall provide information on commercial, academic or other arrangements associated with accessing, collecting or using data for model development, and conflicts of interest (perceived or actual) that might arise from these arrangements.

NOTE: Legislation and guidance can vary greatly. There might be several layers of governance and approval to report in this documentation, across multiple jurisdictions, e.g. research ethics approval, the Data Protection Act 2018, NHS data sharing agreements and commercial collaboration agreements. Where data are sourced internationally, data transfer agreements and frameworks relating to the Data Protection Act 2018, and standard contractual clauses (SCCs) are pertinent.

(BS30440:2023, Section 5.2.2.5)

Collection procedures

- The supplier shall document the names of the organisation(s) involved in data collection, or if this is not possible, provide a justification as to the reasons why *(BS30440:2023, Section 5.2.3.1)*.
- The supplier shall document the mechanisms or procedures used to collect the data (e.g. hardware apparatus or sensor, manual human curation, software program, software application programming interface (API)) *(BS30440:2023, Section 5.2.3.2)*.
- The supplier shall document the variables collected and describe which were used (e.g. structured data: continuous/categorical variables; unstructured data: images, videos, audio). Documentation shall include details of the nature of the information collected, such as:
 - directly observed (e.g. height or blood pressure measurement; raw text);
 - reported by subjects (e.g. survey responses); and
 - indirectly derived from other data (e.g. discretisation of continuous variables, image processing, tokenisation of language, etc.).

(BS30440:2023, Section 5.2.3.3)

- If the data set is labelled for supervised learning and labelling was performed during development of the product (i.e. the data were not pre-labelled), the supplier shall demonstrate that data labelling and feature derivation was done by appropriately qualified and competent staff. The qualifications and experience of such staff shall be made available.

NOTE 1: Frameworks or standards used should be fully cited in the format used in an academic study, or by confirming the relevant standard number, where appropriate, e.g. BS ISO 22857.

NOTE 2: There might be instances where data collected as part of a trial is then linked to other patient data (e.g. magnetic resonance imaging (MRI) acquisition subsequently linked with patient records). These are two separate forms of data for a participant to consent to and consent for both should be presented.

EXAMPLE

A product provider developed an in-house tool to validate the appropriateness of data used to predict the need for a healthcare intervention. The product only allowed clinicians with over 10 years' relevant experience to label data as being fit for the purpose of developing a model. The product enabled clinicians across the world to log on: to understand the provenance and decision making associated with any data item and its label.

(BS30440:2023, Section 5.2.3.4)

Attributes

- The supplier shall document the timeframe and geographical location in which the dataset was created *(BS30440:2023, Section 5.2.4.1)*.

- The supplier shall perform an assessment to verify that the size of the dataset is sufficient to support the intended claims and represent the product user demographic.

EXAMPLE

The provider of AI publishes labels for each of the models created.

Labels state: when a model was created, its data source and the provenance of that data (how it was gathered, from whom, how and why and associated adherence to law) and its validation (who validated it, how and why). The accuracy, sensitivity and specificity of the models was also documented, along with the model terms of use, fitness for purpose and a licensing agreement.

(BS30440:2023, Section 5.2.4.2)

- The supplier shall document if there is information missing from individual records and, if possible, provide a description explaining why it is missing (including, if applicable, the distribution of missing data) or, if this is not possible, provide a justification as to why such a description cannot be provided.

NOTE: Possible distributions of missing data could include missing completely at random (MCAR), missing at random (MAR), or missing not at random (MNAR).

(BS30440:2023, Section 5.2.4.3)

- The supplier shall document and describe sub-populations (e.g. relevant sub-populations and protected characteristic sub-populations associated with the healthcare need) and their respective distributions within the datasets used for training, testing and validation. Where this is not possible, a justification as to why this is not possible shall be provided.

NOTE: Attention is drawn to the Data Protection Act 2018 [3] which sets out protected characteristics, including defining special category data as "Personal data revealing racial or ethnic origin; political opinions; religious or philosophical beliefs; trade union membership; genetic data; biometric data (where used for identification purposes); data concerning health; data concerning a person's sex life; and data concerning a person's sexual orientation".

(BS30440:2023, Section 5.2.4.4)

Handling

- The supplier shall document and justify changes that have been made to the dataset after the data collection process, including data manipulation, data imputation and feature extraction (e.g. discretization of continuous features, part-of-speech tagging, tokenization) (BS30440:2023, Section 5.2.5).

Model

- The supplier documents the process of creating the product and shows that the input and output variables are valid to address the healthcare need (BS30440:2023, Section 5.3).
- The supplier shall document the proposed interaction between end-users and the product, which could include "human-in-command", "human-on-the-loop", "human-in-the-loop" (BS30440:2023, Section 5.3.1.1).
- The supplier shall describe whether the product makes decisions without human oversight and, if so, potential risks resulting from this (BS30440:2023, Section 5.3.1.5).
- The supplier shall document the sensitive data used in the input variable data (e.g. legally protected personal characteristics, genetic/biometric data), and justify its inclusion (BS30440:2023, Section 5.3.2.1.2).

- The supplier shall provide evidence of the validity of the output variable(s) to the healthcare need

NOTE: This might include, but is not limited to:

- a) theoretical formulations;*
- b) technical reports comparing novel methods to established benchmarks/prevaling current methods of measurement; and*
- c) information from the scientific (peer-reviewed) literature.*

EXAMPLE

In the development of a model that needed to predict acute respiratory distress from speech, the research evidencing the statistical validity of using speech as a breathlessness indicator was provided along with additional data relating to the AI methods performance relative to regulators' accepted proxies for evidencing acute respiratory distress.

(BS30440:2023, Section 5.3.2.2.2)

- If surrogate or proxy output variable(s) for the healthcare need are used, the supplier shall provide evidence of the strength of association of the surrogate to the target outcome.

NOTE: Sources of evidence on the strength of association should include:

- a) high level of association: treatment effect on the surrogate end point corresponds to commensurate treatment effect on the final outcomes, as shown in randomized controlled trials;*
- b) consistent association between surrogate end point and final outcomes (typically derived from epidemiological or observational studies); and*
- c) low or emerging evidence of association: biological plausibility of relation between surrogate end point and final outcomes.*

(BS30440:2023, Section 5.3.2.2.3)

- The supplier shall document the AI/ML algorithm(s) used in the model (e.g. name of learning algorithm, model architecture and code packages, where appropriate) *(BS30440:2023, Section 5.3.3.1).*
- The supplier shall document and justify the process of handling missing data, if applicable *(BS30440:2023, Section 5.3.3.2).*
- The supplier shall document and justify the sample size used to train the model *(BS30440:2023, Section 5.3.3.3).*
- If used as part of model development, the supplier shall document the following:
 - data pre-processing procedure(s);
 - hyperparameter selection, optimisation and thresholding; and
 - internal validation procedure(s).

(BS30440:2023, Section 5.3.3.5)

- The supplier shall document and justify the use of any decision thresholds that convert probabilities into discrete outputs in the final model *(BS30440:2023, Section 5.3.4.2).*

Testing

- The supplier shall document the performance of the product using appropriate metrics for the proposed task using testing data (e.g. accuracy, sensitivity, specificity, positive predictive value, negative predictive value, area under the curve, calibration, etc. for classification tasks) *(BS30440:2023, Section 5.3.5.2).*

- The supplier shall document and justify the choice of performance and safety metrics used as part of the model testing (BS30440:2023, Section 5.3.5.3).
- The supplier shall document exploration for bias undertaken in model training and testing and document any bias mitigation strategies introduced (BS30440:2023, Section 5.3.5.4).
- The supplier shall document, where the product interfaces with a human, that its interface and behaviour are consistent with user expectation and predictable for the user.

NOTE: Products using machine learning should be sensitive to nuances in the context or setting. This should not lead to behaviours that are unexpected by the user, and which are disruptive or confusing. Examples include changes in lighting or background noise, which might not be regarded as contextually relevant to the task by the user.

(BS30440:2023, Section 5.4.2)

- The supplier shall identify and meet user training needs, verifying that users have completed training on:
 - what the product should do;
 - how well it should satisfy its intended use; and
 - what the product's operational design domain boundaries are.

NOTE: Training for interaction with products is likely to go beyond the scope of training traditionally provided for the use of (physical) medical devices or standard IT products. In addition to being able to operate the product, the user also needs to understand the applicability, relevance and limitations of probabilistic approaches and outputs for their tasks, and users need to know how to identify potentially inadequate or inappropriate outputs and behaviours, which might not result from a malfunction (as is the case with traditional medical devices) but which could arise from mismatches in training data and operational realities.

(BS30440:2023, Section 5.4.3)

Control

- The supplier shall make arrangements for users to take control from the product and to override product decisions. The supplier shall identify user requirements for overriding the product and demonstrate that risks associated with this transition from autonomous to manual mode of operation have been mitigated.

NOTE: There are few, if any, examples of truly autonomous systems that operate without human oversight. Therefore, the role of users and their requirements need to be determined explicitly, including the transition from supervisory and monitoring functions to manual control.

(BS30440:2023, Section 5.4.5)

- The supplier shall carry out a risk assessment as to the potential for automation bias and over-reliance on the AI and demonstrate that appropriate risk controls have been implemented.

NOTE: Automation bias (or automation-induced complacency) refers to a breakdown in the interaction between the user and the automation, where the user relies on the automation uncritically. Strategies for reducing the likelihood of automation bias include interface design, the use of explanation and feedback, training and an open and just culture that enables users to disagree with AI decisions without fear of repercussions. However, even with these strategies, automation bias can and does occur at times. In the risk assessment, the potential consequences of over-reliance on the AI should be identified and risk controls for mitigating the severity of the consequences implemented.

(BS30440:2023, Section 5.4.6)

Carbon Impact

- The supplier shall document an estimate of the carbon impact incurred during training and execution. This shall be achieved using carbon intensity trackers (*BS30440:2023, Section 5.5.1*).

Healthcare effectiveness

- The supplier shall provide evidence of the effectiveness and risk of the product in addressing a healthcare need in a relevant real-world setting.

NOTE: The NICE Evidence standards framework for digital health technologies provides guidance on sufficient evidence standards for digital health technologies.

(BS30440:2023, Section 6.1.1)

- The supplier shall provide details of procedures for detecting adverse events during healthcare evaluations of the product (*BS30440:2023, Section 6.1.4*).
- The supplier shall document details of cases where, during the validation phase, performance of the product in a real-world setting was inadequate, including how these instances were detected, and subsequent remedial actions taken (*BS30440:2023, Section 6.1.5*).
- The supplier shall perform sub-population analysis (specifically: healthcare relevant sub-populations, and sub-populations with protected characteristics) to show performance in real world settings (*BS30440:2023, Section 6.1.6*).
- The supplier shall provide metrics on how the product performs with incomplete data and the effect on performance (*BS30440:2023, Section 6.1.7*).

External validity

- The supplier shall provide evidence of the impact of novel settings on the performance (including, but not limited to, safety profile and effectiveness) of the product.

NOTE: Novel setting could include different time periods, geographical and organisational contexts. The chosen settings should be relevant to the intended use of the product.

(BS30440:2023, Section 6.2.1)

- Evidence shall include technical reports or published papers describing the application of the product, in accordance with relevant standards for evaluation of products in healthcare settings (*BS30440:2023, Section 6.2.2*).

Equity and bias

- The supplier shall document how it and its suppliers accommodate the diversity of users impacted by the product (*BS30440:2023, Section 6.3.1.1*).
- The supplier shall undertake a risk assessment to identify risks of bias in model development and deployment that might result in inequitable outcomes.
 - The risk assessment shall document the degree of potential impact and harm for sub-groups and outline mitigations.
 - The risk assessment shall include:
 - a) explanations for the mechanism for monitoring of real-world impact on sub-groups post-deployment; and

- b) details of when the assessment is to be reviewed, e.g. at key stages of model maintenance, new use-cases, updates and decommissioning.
(BS30440:2023, Section 6.3.2.1-3)

Resourcing

- The supplier shall provide a description of the costs, resources and staff/user training needed to deploy and maintain the product in real world settings, including but not limited to:
 - a) computation, information technology and network requirements;
 - b) input data parameter requirements for successful operation in the deploying environment; and
 - c) staffing and training requirements.

(BS30440:2023, Section 7.1.1)

- The supplier shall perform a risk assessment of the deployment of the product in relevant operational scenarios. The assessment shall make clear who the risk affects (e.g. patient, user, organisation, etc.) using an appropriate method and associated mitigations.

NOTE: DCB0160 is an example of an appropriate method and associated mitigations, used to risk assess the deployment of healthcare systems. It evidences technical risks and their co-dependency in healthcare, and associated mitigations.

(BS30440:2023, Section 7.1.2)

- The supplier shall document the expectations of ongoing product support and maintenance for both it and the adopter (BS30440:2023, Section 7.1.3).
- The supplier shall document the process for agreeing a standard operating procedure (SOP) between the supplier and adopter. This process shall include plans for the regular review of the SOP.

NOTE: The SOP should incorporate as a minimum:

- a) an agreed format and scope;
- b) a list of personnel from both the supplier and adopter organisations, detailing roles and responsibilities;
- c) activities each personnel member is tasked with undertaking; and
- d) a direct link to risk management activities.

(BS30440:2023, Section 7.1.4)

- The supplier shall make available documentation stating the anticipated cost and resource impact to the healthcare system associated with using the product within the relevant healthcare context, including resources relating to (but not limited to):
 - a) infrastructure;
 - b) operations;
 - c) staffing;
 - d) monitoring;
 - e) training; and

- f) expected healthcare outcomes.

(BS30440:2023, Section 7.2.1)

- The supplier shall have in place a risk management process to identify risks and mitigations both during development and deployment (also known as DCB0129). Risk assessments shall be carried out to address the following:
 - a) situations where the product itself might lead to outcomes that are unsafe or undesirable for users;
 - b) situations where the product might be used to facilitate or enable abuse, particularly vulnerable groups;
 - c) potential interaction failure, i.e. situations where users might inadvertently (but not maliciously) misuse the product, leading to outcomes that are unsafe or undesirable for users; and
 - d) the impact of outlier input data (e.g. if infant ECG data were the input for a product trained on adult ECG data).

NOTE 1: Risk management standards, BS EN ISO 14971 and DCB0129, should be used to assist in conforming to this subclause. DCB0129 requires evidence to be output in the form of a safety case and a hazard log. Similarly, BS EN ISO 14971 requires evidence to be output to a risk management file.

NOTE 2: Risk management processes could follow those outlined by the DCB0129 and DCB0160. (BS30440:2023, Section 7.4.1)

- Information about AI Models or algorithms shall be written in accessible language and made publicly available such that an individual can query outcomes to understand how a decision was reached.
- The supplier shall document the process for alerting users to potential safety issues *(BS30440:2023, Section 8.1.1.4)*.
- The supplier shall document the post-deployment plan for routine monitoring of product performance and safety schedule (i.e. post-market surveillance).

NOTE: This might include, but is not limited to:

- a) *the healthcare context, to verify there have been no changes that would affect the performance of the product;*
- a) *the input and output variables to maintain consistent performance over time;*
- b) *performance in different sub-populations to detect the emergence of algorithmic bias;*
- c) *usage and user experience so that the system continues to be used as expected, and to identify training requirements; and*
- d) *patient alerts, software updates, instructions for use changes, patient review and device recalls.*

(BS30440:2023, Section 8.1.2)

- Suppliers shall provide a mechanism to monitor drift and identify selective underperformance in specific groups and sub-populations (i.e. statistical bias). The mechanism shall be transparent and accessible to users of the product. The mechanism shall include details of reporting pathways for both the supplier and user.

NOTE: Monitoring for bias and drift should be via analytics dashboards with potentially automated warnings. However, other factors, such as operating environment, could impact the performance of the model.

(BS30440:2023, Section 8.1.3.1)

- The supplier shall document processes to maintain a traceable record of modifications to the model (e.g. model updates) and non-model parts of the product (e.g. bug fixes) in line with good practice in software development including the reason for change, any impact on stakeholders and any actions required for stakeholders *(BS30440:2023, Section 8.2.1-3)*
- The supplier shall document processes for decommissioning the product, including considering data ownership and storage *(BS30440:2023, Section 8.3)*.